

Insurance Claims & Risk Analysis

Executive Summary

Objective

The objective of this analysis was to identify factors associated with higher-cost insurance claims and longer claim resolution times.

Data & Methodology

The analysis used the Minitab Insurance Fraud Dataset, supplemented with a U.S. ZIP code reference table for geographic enrichment.

The source data was imported, validated, and cleaned in PostgreSQL through DBeaver. Quality checks included row-count validation, duplicate checks, missing-value handling, range checks, date validation, and data-type review. The cleaned dataset was then enriched with geographic region information and used to create an SQL view for analysis.

The geographic join matched approximately 76% of claims. Investigation determined that the lower match rate was due to incomplete coverage in the free ZIP code reference dataset, rather than an issue with the join logic. Unmatched records were retained and treated as unknown geographic classifications.

Claim Cost

Key Findings

- Average claim amounts were relatively consistent across vehicle category, vehicle value, safety-rating groups, previous claim history, and region.
- There was variation in claim frequency across accident sites, with local accidents representing a much larger share of the claims than highway accidents. However, average claim cost did not vary substantially across accident sites.

Overall claim severity varied substantially despite the limited differences between group averages. Claim amounts ranged from approximately \$2.6K to \$101.8K, with an average of \$22.9K and a median of \$20.5K.

A review of the ten highest-cost claims found more large vehicles among the most expensive claims, but high claim severity occurred across a broad range of vehicle values, accident locations, and regions. No single characteristic clearly explained the highest-cost claims.

Overall finding: The analysis did not identify one dominant individual characteristic associated with higher claim severity. The substantial variation in overall claim amounts suggests that combinations of characteristics may be worth investigating further.

Claim Resolution Time

Key Findings

- Average days open showed little variation across accident site, fraud status, form defects, and region.
- Claim channel showed the most notable variation: online claims averaged 4.00 days open, compared with 9.02 days for phone claims and 10.02 days for broker claims.
- Broker and phone claims represented larger shares of the dataset than online claims, so further analysis would be needed to determine whether the difference is attributable to the submission channel itself or to other characteristics of those claims.

Overall finding: Resolution times were generally consistent across the factors analyzed, with claim channel emerging as an area for further investigation.

Recommendations & Further Investigation

Based on the findings, several areas could be explored further:

1. Examine combinations of factors.

Since individual characteristics showed limited relationships with claim severity, multi-factor analysis could possibly identify combinations associated with higher-cost claims or longer resolution times.

2. Investigate claim-channel differences.

The shorter resolution time for online claims could indicate differences in workflow or claim complexity. Comparing claim characteristics across channels could help to determine what is driving the difference.

3. Expand the available data.

Additional policyholder, coverage, operational, and exposure data could provide greater insight into factors influencing claim cost and resolution time.

4. Improve geographic coverage.

A more complete ZIP-code reference dataset would allow more comprehensive regional analysis.

Conclusion

The analysis did not identify a single characteristic that consistently explained higher claim costs or longer resolution times. Overall claim amounts varied substantially, while the average claim costs across the individual factors analyzed remained relatively similar.

These findings suggest that higher-cost claims may involve combinations of characteristics. Claim-processing differences by submission channel may also be worth investigating further.

The analysis also demonstrates the importance of validating and documenting data quality before drawing business conclusions. Known limitations, including incomplete geographic enrichment, were retained and documented rather than overlooked.